Model Accuracy

What do you know about Machine Learning?
- The basic idea:
- **Model types:**
- Prediction Accuracy vs Model Interpretability

Accuracy of Regression (Parametric)
- Training vs Test MSE
- Bias vs Variance tradeoff

Accuracy of Classification
- Bayes Classifier
- K-Nearest Neighbour

What do you know about Machine Learning?

The basic idea:

$$Y = f(X) + e$$

Assume that there is a function relating X and Y, where e is the error term from random fluctuations in data

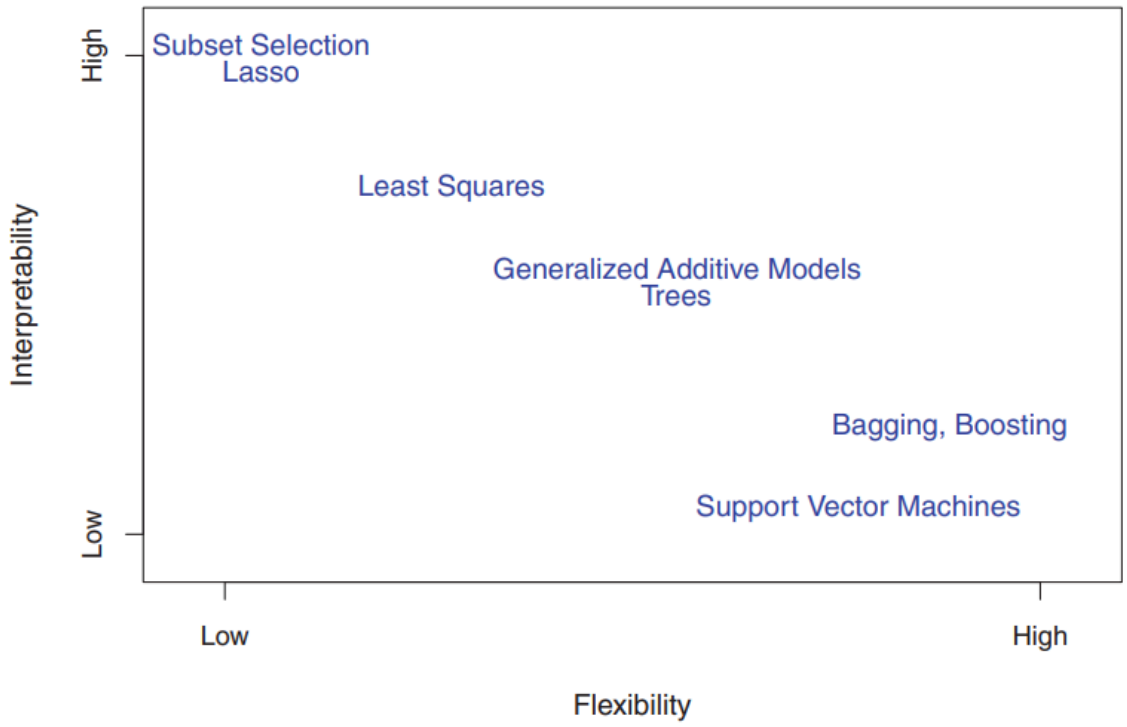
Model types:

- Predictive Models
  - To get an output closest to observed data
  - Does not concern the form of the unknown function, as long as prediction is accurate
  - E.g) *Stock market prediction*
- Inference Models
  - For visualising a few parameter's relation with the output
  - For Inference, the nature of the unknown function must be well-known
  - E.g) A dataset consisting of salary, age, level of education, country. There can be a few questions:

- *Which parameters affect the salary?*
- *How does each predictor relate to the output? Is it a positive or negative correlation? Strong or weak?*
- *Can the relationship be expressed in linear form? Or is it more complicated?*

Prediction Accuracy vs Model Interpretability

When the model can fit every data point (flexible), it tells very little about the relationship between each parameters and the output



Accuracy of Regression (Parametric)

Training vs Test MSE

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$$

- MSE is the *mean squared difference* between observed data and predicted output
- 2 types of MSE:
  - Training MSE (MSE between training data and model prediction)
    - Decreases as flexibility increases
  - Test MSE (MSE between new data and model prediction)
    - U-shaped, increased flexibility beyond a point makes it too susceptible to noise
- Different data structures retains the general characteristics of training and test MSE

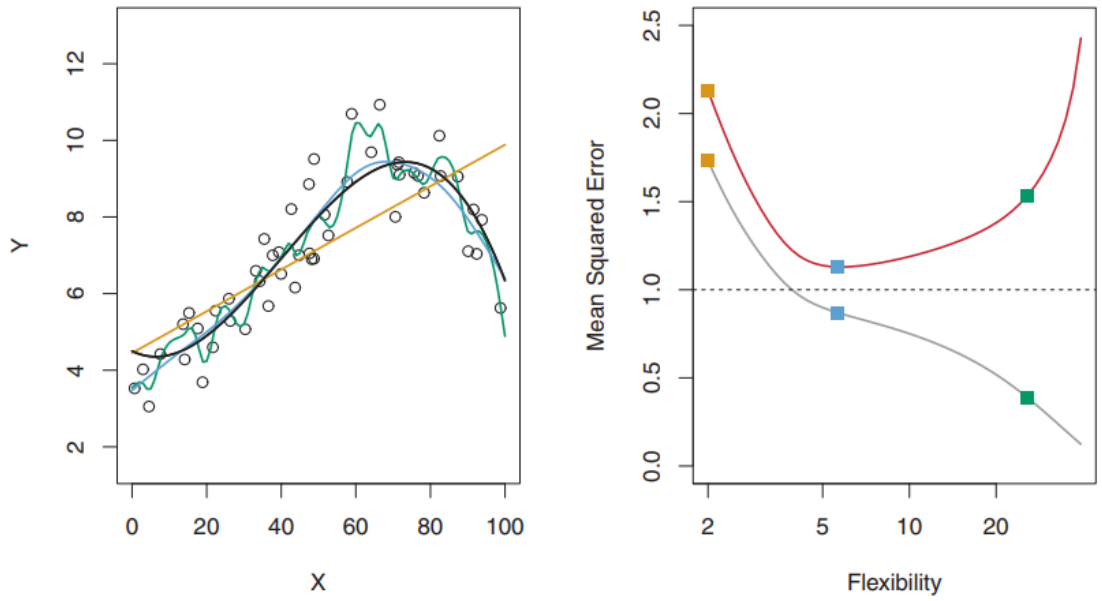
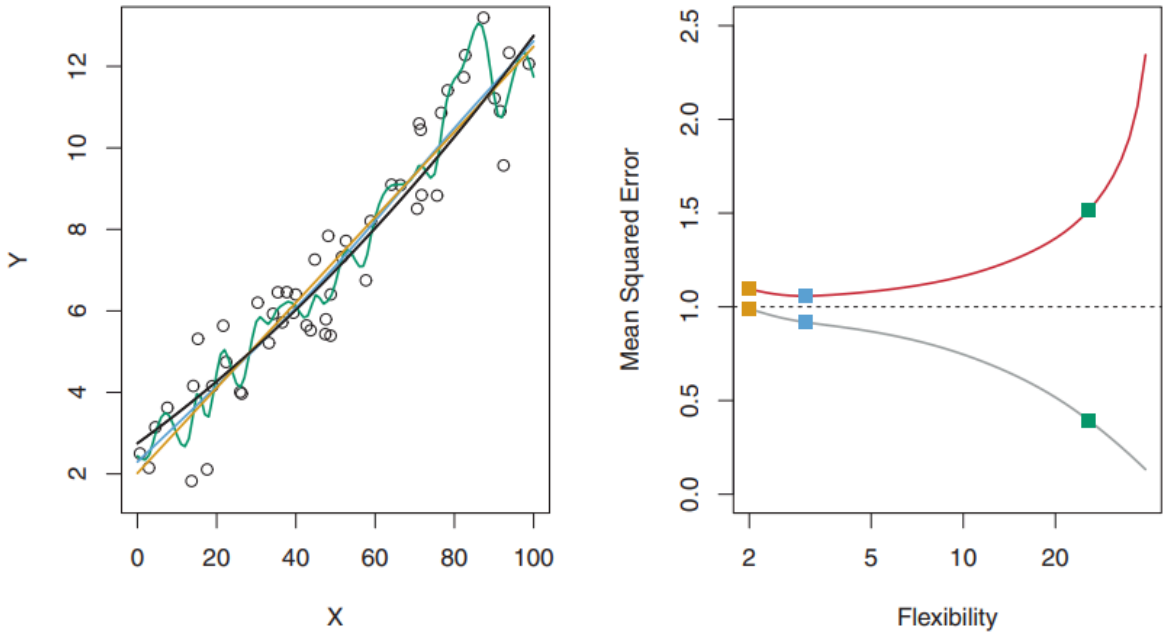
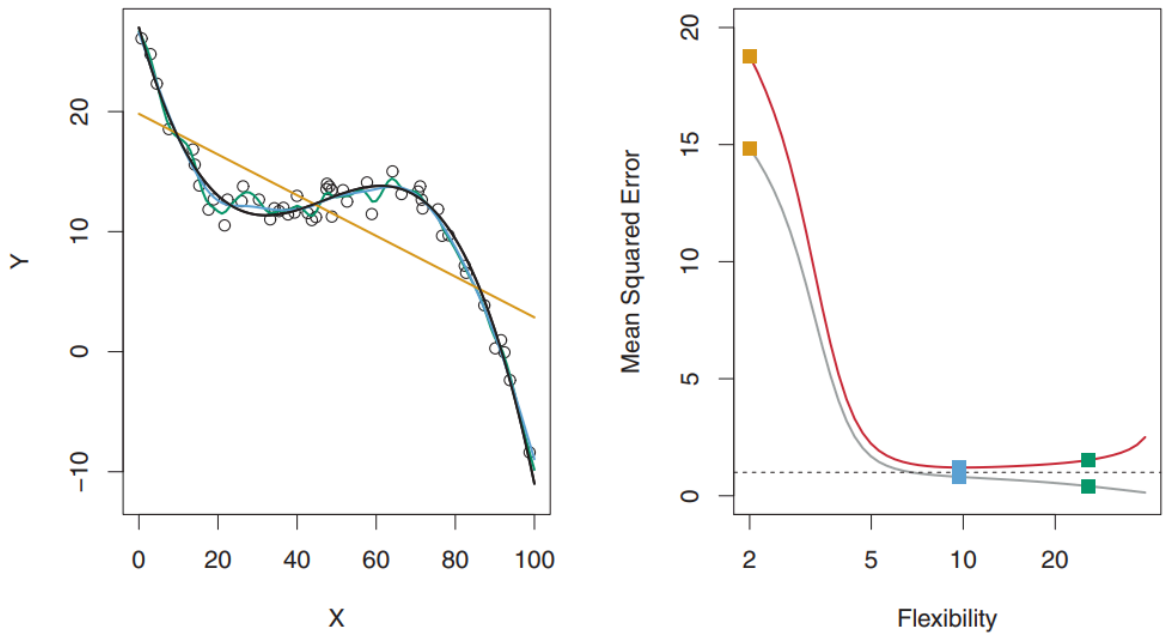


FIGURE 2.9. Left: *Data simulated from f, shown in black. Three estimates of f are shown: the linear regression line (orange curve), and two smoothing spline fits (blue and green curves).* Right: *Training MSE (grey curve), test MSE (red curve), and minimum possible test MSE over all methods (dashed line). Squares represent the training and test MSEs for the three fits shown in the left-hand panel.*





Figures above: *3 different data nature, using different regression techniques: linear regression, spline fitting*

Bias vs Variance tradeoff

$$E\left[\left(y_0 - \hat{f}(x_0)\right)^2\right] = \mathrm{Var}\left(\hat{f}(x_0)\right) + \left[\mathrm{Bias}\left(\hat{f}(x_0)\right)\right]^2 + \mathrm{Var}(\varepsilon)$$

The *Expected Test Mean Squared Error* has 3 components:

- Variance of the estimated function (flexibility of the model)
- Bias of function squared (how forcefully the sample is fit into an assumption)
- Variance of the irreducible error

▼ Derivation see below:

$y_0 = f(x_0) + \varepsilon$

$E[(y_0 - \hat{f}(x_0))]^2 = E[(f(x_0) - \hat{f}(x_0)) + \varepsilon]^2$

<u>Add and subtract and let f(x) = f</u>:

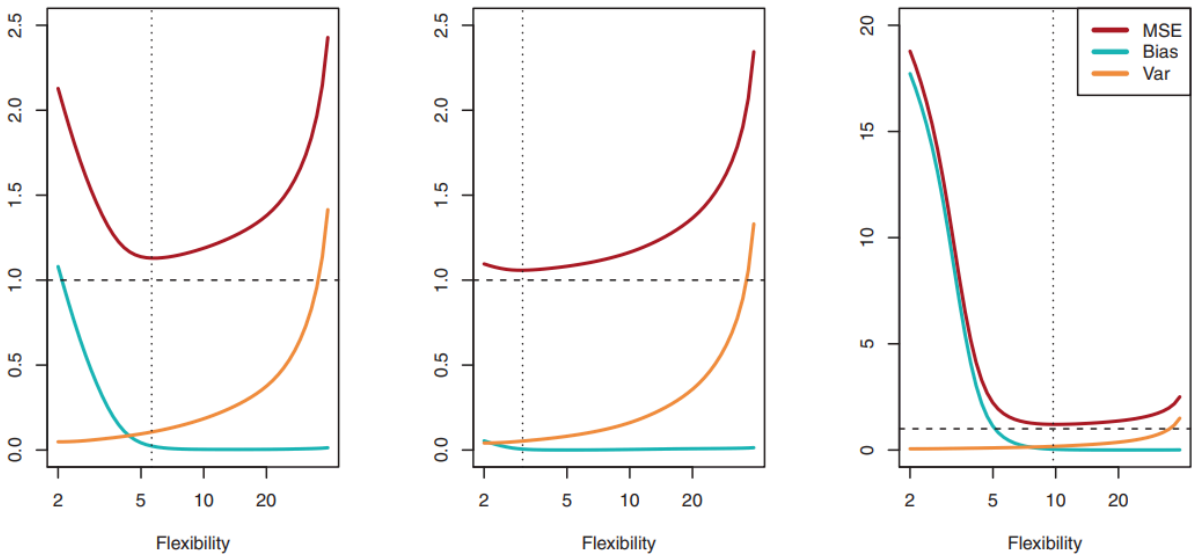
$E[(f - E[\hat{f}]) + (E[\hat{f}] - \hat{f}) + \varepsilon]^2$

Since $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$, the eqn above becomes

- $E[(f - E[\hat{f}])^2] = (f - E[\hat{f}])^2$
  - This is the **Bias squared**
- $E[E[\hat{f}] - \hat{f}]^2 = Var(\hat{f}(\hat{x}_0))$
- $E[\varepsilon^2] = Var(\varepsilon)$

- For terms 2ab, 2bc,2ac: Since $E[\varepsilon] = 0$ and $E[E[\hat{f}] - \hat{f}] = E[\hat{f}] - E[\hat{f}] = 0$, they are omitted

Minimum E[MSE] can be obtained by minimising both Var(f_hat) and Bias



(Continued from the figures above, min MSE is obtained from vertical dotted line)

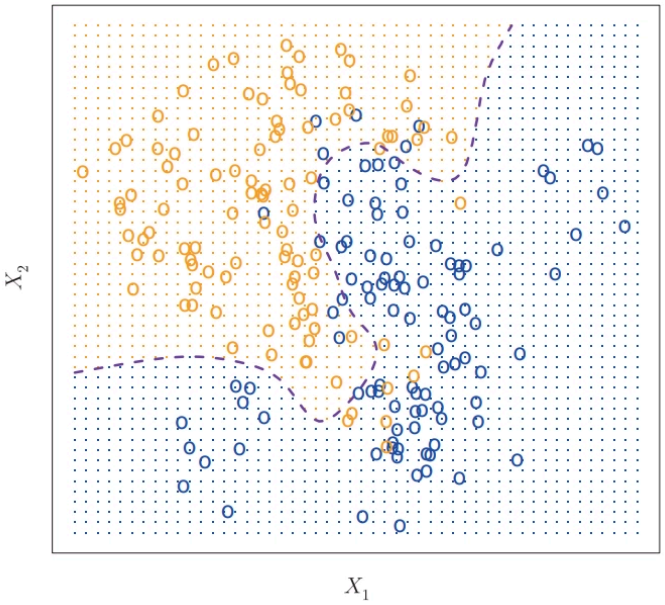
Accuracy of Classification

Bayes Classifier

$$\text{Ave}\left(I\left(y_0 \neq \hat{y}_0\right)\right)$$

Same as MSE, *test error rate* is the rate at which predicted observation ≠ test observation

- *I(y...≠ ...)* is a function that returns 1 if statement is True, and 0 if False
- This formula test error rate is linked to Bayes Classifier



The circles are training observations. For each observation, we can classify the circle as orange or blue using the *Bayes Classifier,* where

$Pr(Y = 1|X = x_0) > 0.5$ represents orange.

To classify things, we want to draw a line — the *Bayes decision boundary* — such that predicted observations is classified as orange / blue depending on which region they fall on. Since there's overlapping between blue and orange classes, there is an irreducible error, the *Bayes error rate.* The goal is to produce a decision boundary that yields the minimum error rate, which is the *Bayes error rate.*

- $1 - E(\max_j \Pr(Y = j \mid X)) > 0$ cuz there's overlapping

So, when conditional probabilities are unknown, how do we estimate this optimal Bayes error rate?

K-Nearest Neighbour

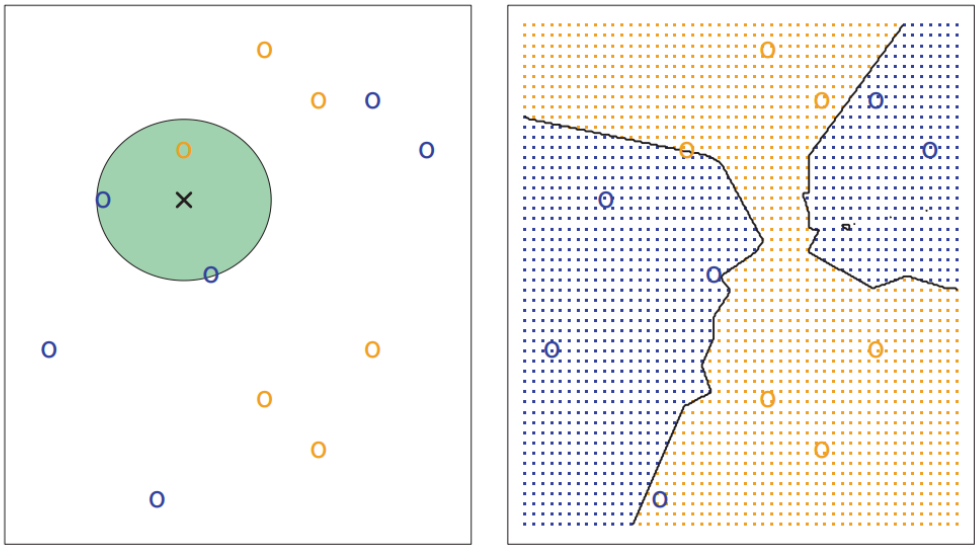


FIGURE 2.14. *The KNN approach, using $K = 3$, is illustrated in a simple situation with six blue observations and six orange observations.* Left: *a test observation at which a predicted class label is desired is shown as a black cross. The three closest points to the test observation are identified, and it is predicted that the test observation belongs to the most commonly-occurring class, in this case blue.* Right: *The KNN decision boundary for this example is shown in black. The*

- Class of 'x' is estimated based on the highest probability of K nearest training points

  ▼ $\Pr(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$

  This is just a sum of the indicator function "*I(y=j)*" results of K nearest points and divided by K

- Repeat this procedure for every possible X value to get the decision boundary

The K-value affects the flexibility of the boundary. The lower the K, the more flexible the decision boundary, but the more susceptible it is to noise.

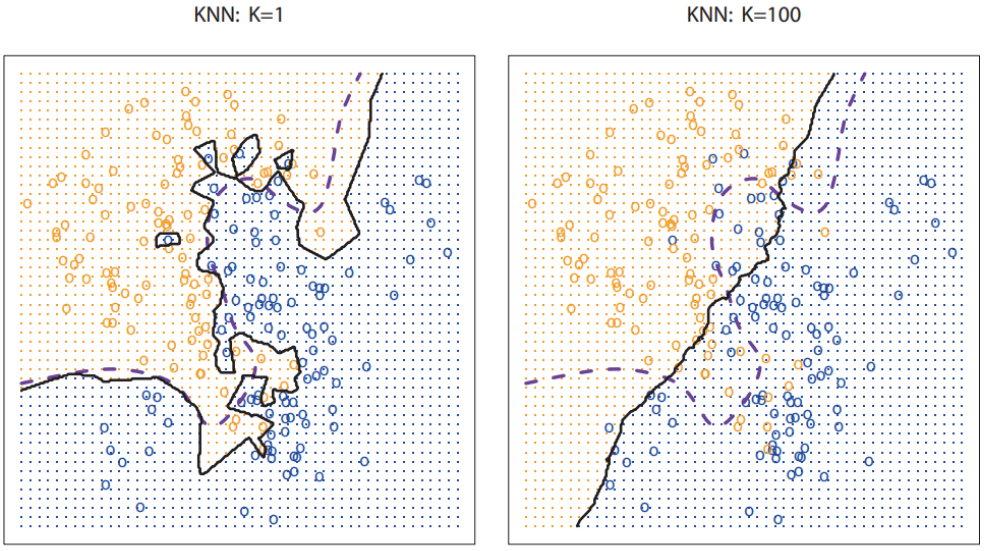


FIGURE 2.16. *A comparison of the KNN decision boundaries (solid black curves) obtained using $K = 1$ and $K = 100$ on the data from Figure 2.13. With $K = 1$, the decision boundary is overly flexible, while with $K = 100$ it is not sufficiently flexible. The Bayes decision boundary is shown as a purple dashed line.*